

## ✓ SIMPLE REGRESSION ANALYSIS

Marketing ROI analysis using TV, Radio and Social Media as features. dependent variable/ target variable is Sales, while the independent variables/features are TV, Radio and Social Media. The goal of this project is to recommend the marketing channel with the strongest ROI impact

### import libraries and packages

**pandas** for data manipulation

**statsmodels** for statistical modeling

**seaborn** for plotting

## ✓ Import packages

import pandas as pd import numpy as np import seaborn as sns

```
# Load dataset
import pandas as pd
marketing = pd.read_csv("marketing.csv")

# Examine first 10 rows of dataset
print(marketing.head(10))
```

|   | TV   | Radio     | Social_Media | Sales      |
|---|------|-----------|--------------|------------|
| 0 | 16.0 | 6.566231  | 2.907983     | 54.732757  |
| 1 | 13.0 | 9.237765  | 2.409567     | 46.677897  |
| 2 | 41.0 | 15.886446 | 2.913410     | 150.177829 |
| 3 | 83.0 | 30.020028 | 6.922304     | 298.246340 |
| 4 | 15.0 | 8.437408  | 1.405998     | 56.594181  |
| 5 | 29.0 | 9.614382  | 1.027163     | 105.889148 |
| 6 | 55.0 | 24.893811 | 4.273602     | 198.679825 |
| 7 | 31.0 | 17.355042 | 2.289855     | 108.733932 |
| 8 | 76.0 | 24.648898 | 7.130116     | 270.189400 |
| 9 | 13.0 | 0.431128  | 2.229423     | 48.280582  |

```
#examine the shape of the data
print(marketing.shape)
```

```
(4572, 4)
```

```
#examine the size of the data
print(marketing.size)
```

```
18288
```

```
print(marketing.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
TV              4572 non-null    float64
Radio           4572 non-null    float64
Social_Media    4572 non-null    float64
Sales           4572 non-null    float64
```

```

0   TV          4562 non-null   float64
1   Radio       4568 non-null   float64
2   Social_Media 4566 non-null   float64
3   Sales       4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
None

```

```
print(marketing.describe())
```

|       | TV          | Radio       | Social_Media | Sales       |
|-------|-------------|-------------|--------------|-------------|
| count | 4562.000000 | 4568.000000 | 4566.000000  | 4566.000000 |
| mean  | 54.066857   | 18.160356   | 3.323956     | 192.466602  |
| std   | 26.125054   | 9.676958    | 2.212670     | 93.133092   |
| min   | 10.000000   | 0.000684    | 0.000031     | 31.199409   |
| 25%   | 32.000000   | 10.525957   | 1.527849     | 112.322882  |
| 50%   | 53.000000   | 17.859513   | 3.055565     | 189.231172  |
| 75%   | 77.000000   | 25.649730   | 4.807558     | 272.507922  |
| max   | 100.000000  | 48.871161   | 13.981662    | 364.079751  |

```
print(marketing.columns)
```

```
Index(['TV', 'Radio', 'Social_Media', 'Sales'], dtype='object')
```

## ✓ DATA CLEANING

```
print('missing values:\n',marketing.isnull().sum())
```

```

missing values:
TV          10
Radio        4
Social_Media 6
Sales        6
dtype: int64

```

## ✓ drop all missing values in datasets

```
print(marketing.dropna(inplace=True))
```

```
None
```

```
print(marketing.isnull().sum())
```

```

TV          0
Radio        0
Social_Media 0
Sales        0
dtype: int64

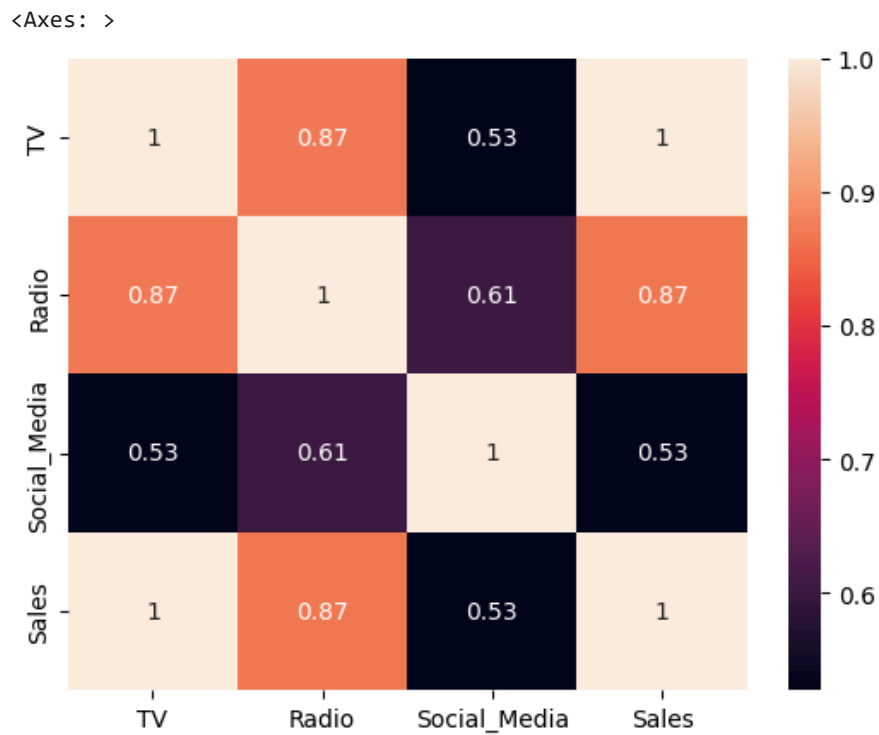
```

## ✓ Exploratory data analysis

```

import seaborn as sns
# create a heatmap of the dataset
sns.heatmap(marketing.corr(), annot=True)

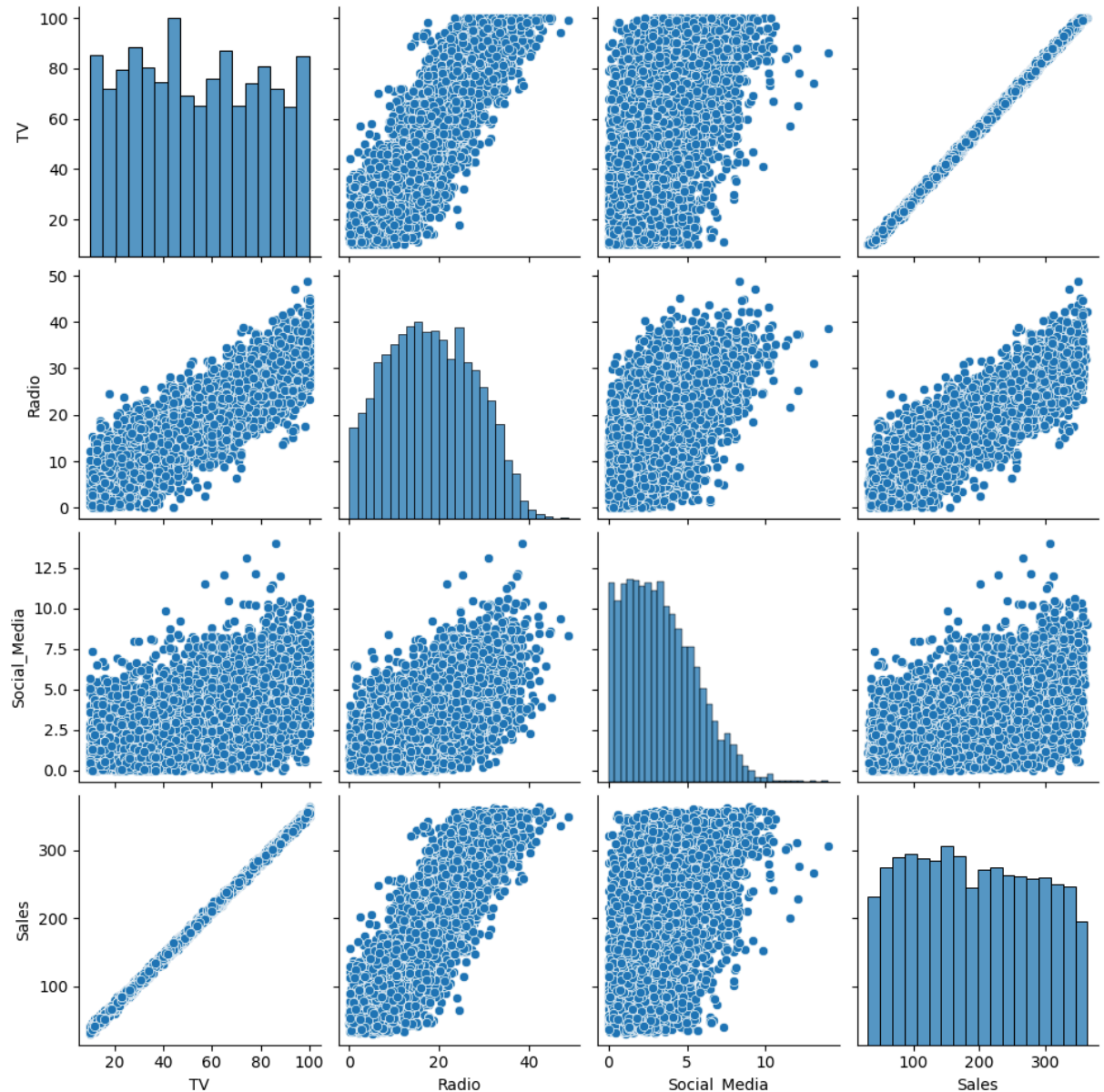
```



**from the heatmap** This gives a visual explanation of the correlation between TV, Social\_Media and Radio. TV has an almost perfect correlation with Sales TV has a higher impact on sales

```
# Create pairwise scatterplots of data set
import seaborn as sns
sns.pairplot(marketing)
```

```
<seaborn.axisgrid.PairGrid at 0x7e7a59cc9bb0>
```



From the scatterplot matrix, you can observe a few linear relationships:

- TV and Sales ( almost perfectly linear)
- TV and Radio
- Radio and Sales
- Social Media and Radio

## ✓ Model development

Based on the above scatterplots, we could probably run a simple linear regression on any of the three relationships identified. For this project our focus will be on the relationship between TV and Sales as they are the most linearly correlated

To do this, we first subset the variables of interest from the dataframe by using double square brackets `[[ ]]`, and listing the names of the columns of interest.

```
# Subset Data
```

```
ols_data = marketing[['TV', 'Sales']]
```

```
print(ols_data)
```

```

      TV      Sales
0   16.0   54.732757
1   13.0   46.677897
2   41.0  150.177829
3   83.0  298.246340
4   15.0   56.594181
...   ...   ...
4567  26.0   94.685866
4568  71.0  249.101915
4569  44.0  163.631457
4570  71.0  253.610411
4571  42.0  148.202414

```

```
[4546 rows x 2 columns]
```

Next, we can construct the linear regression formula

- Dependent variable y = Sales
- Independent Variable X = TV

```
# Write out formula
print('ols_formula = Sales ~ TV')
```

```
ols_formula = Sales ~ TV
```

$y = mx + c + e$

**Let's build the simple linear regression model in statsmodels**

```
# Import ols function
from statsmodels.formula.api import ols
```

```
# Build OLS, fit model to data
ols_formula = 'Sales ~ TV'
ols_data = marketing[['TV', 'Sales']]
OLS = ols(ols_formula, ols_data)
model = OLS.fit()
```

```
print(ols_data.head())
```

```

      TV      Sales
0   16.0   54.732757
1   13.0   46.677897
2   41.0  150.177829
3   83.0  298.246340
4   15.0   56.594181

```

Start coding or [generate](#) with AI.

Let's get the summary of the model, coefficients and more statistics about the model. this helps us evaluate the model and interpret the results

```
print( model.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          Sales      R-squared:                0.999
Model:                  OLS       Adj. R-squared:           0.999
Method:                 Least Squares   F-statistic:            4.517e+06
Date:                  Thu, 11 Jun 2026   Prob (F-statistic):      0.00
Time:                  15:17:49         Log-Likelihood:         -11366.
No. Observations:      4546           AIC:                   2.274e+04
Df Residuals:          4544           BIC:                   2.275e+04
Df Model:              1
Covariance Type:       nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept      -0.1325      0.101      -1.317      0.188      -0.330      0.065
TV              3.5615      0.002     2125.272      0.000      3.558      3.565
=====
Omnibus:                0.052   Durbin-Watson:           1.998
Prob(Omnibus):          0.974   Jarque-Bera (JB):         0.031
Skew:                  -0.001   Prob(JB):                 0.985
Kurtosis:              3.012   Cond. No.                 138.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
print(f'R-Squared = {model.rsquared:.3f} means 99.9% of Sales variation is explained by the
```

```
R-Squared = 0.999 means 99.9% of Sales variation is explained by the model
```

```
print(f'Equation: Sales = 3.5615 * TV + (-0.1325)')
```

```
Equation: Sales = 3.5615 * TV + (-0.1325)
```

### Linear equation

sales = 3.5615 \* TV + 3.6116

## FROM THE OLS SUMMARY

**R-Squared: this indicates that TV explains 99% of Sales Variation**

**TV coef: 3.5615** what this means is that for every 1 unit increase in TV, Sales go up by 3.56 units (i.e 3.5X ROI)

**P-values,**

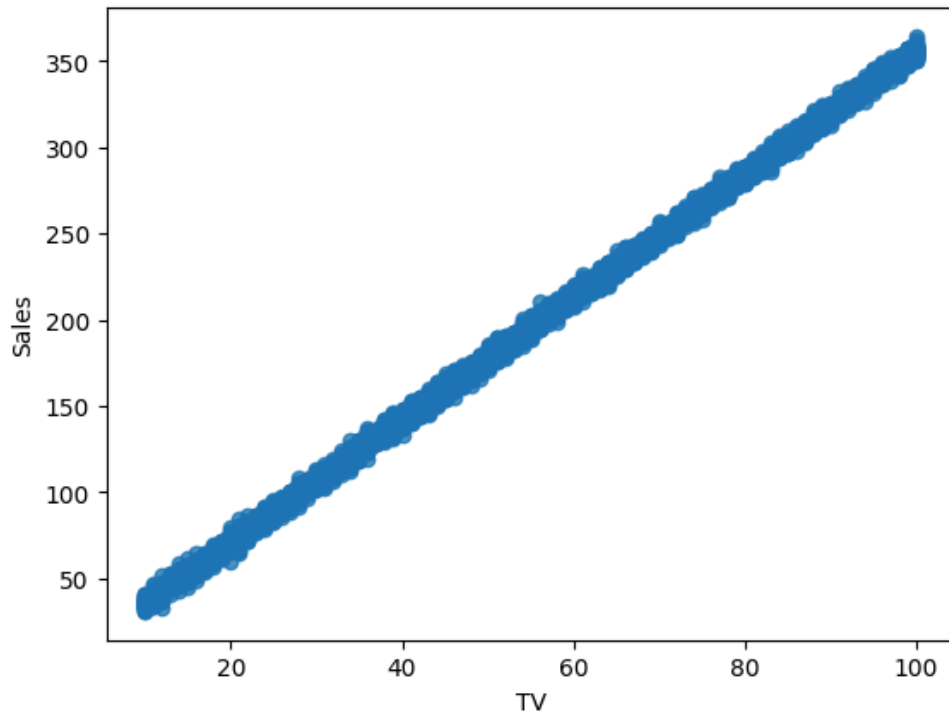
**P|t| for TV is 0.00**, so TV is statistically significant, as  $p|t| < 0.00$  shows a feature is useful// significant while  $> 0.00$  signals noise

**DF model = 1**, tells us that this is a simple regression model as only one feature was used

## ✓ visual of the regression line.

```
import seaborn as sns
sns.regplot(x = "TV", y = "Sales", data = ols_data)
```

<Axes: xlabel='TV', ylabel='Sales'>



## ✓ model assumptions

1. Linearity
2. Normality
3. Independent observations
4. Homoscedasticity

we already checked the linearity assumption by creating the scatterplot matrix and we observed that TV and Sales assume linearity

The normality and homoscedasticity assumptions focus on the distribution of errors. Thus, we can only check these assumptions after we have constructed the model. To check these assumptions, we will check the residuals, as an approximation of the errors.

To more easily check the model assumptions and create relevant visualizations.

```
X = ols_data["TV"]
X.head()
```

|   | TV   |
|---|------|
| 0 | 16.0 |
| 1 | 13.0 |
| 2 | 41.0 |
| 3 | 83.0 |
| 4 | 15.0 |

**dtype:** float64

```
# Get predictions from model
fitted_values = model.predict(X)
```

```
print(fitted_values)
```

```
0      56.851733
1      46.167191
2     145.889585
3     295.473177
4      53.290219
...
4567    92.466874
4568   252.735008
4569   156.574128
4570   252.735008
4571   149.451099
Length: 4546, dtype: float64
```

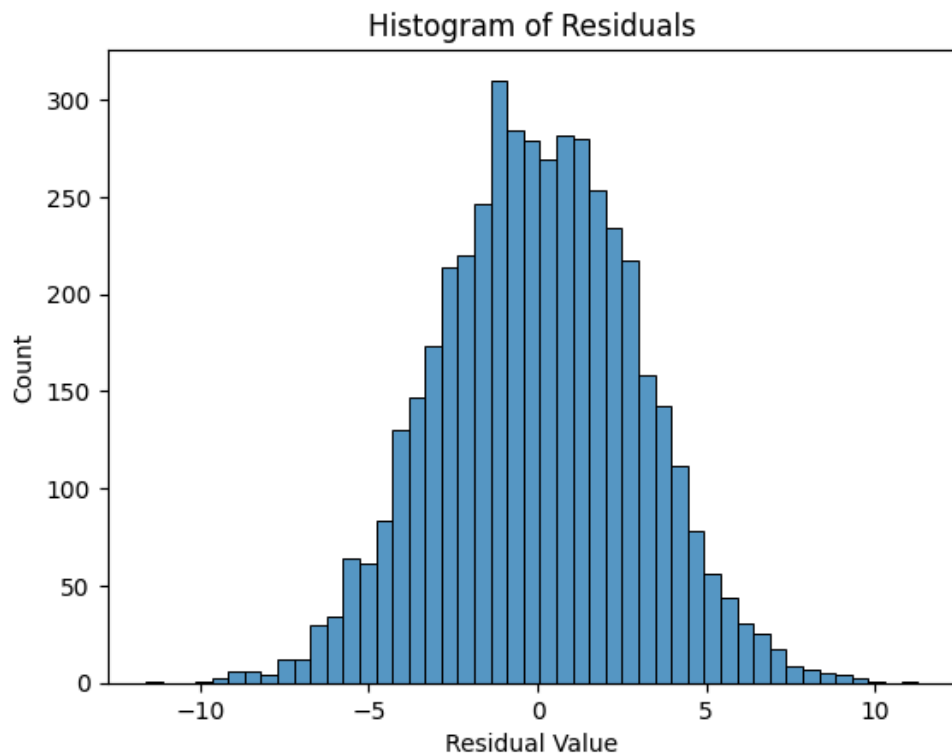
```
# Calculate residuals
residuals = model.resid
```

## ✓ Check the normality assumption

From the below histogram, you may notice that the residuals are almost normally distributed. In this case, it is likely close enough that the assumption is met.

```
import matplotlib.pyplot as plt
fig = sns.histplot(residuals)
fig.set_xlabel("Residual Value")
fig.set_title("Histogram of Residuals")
plt.show()
```

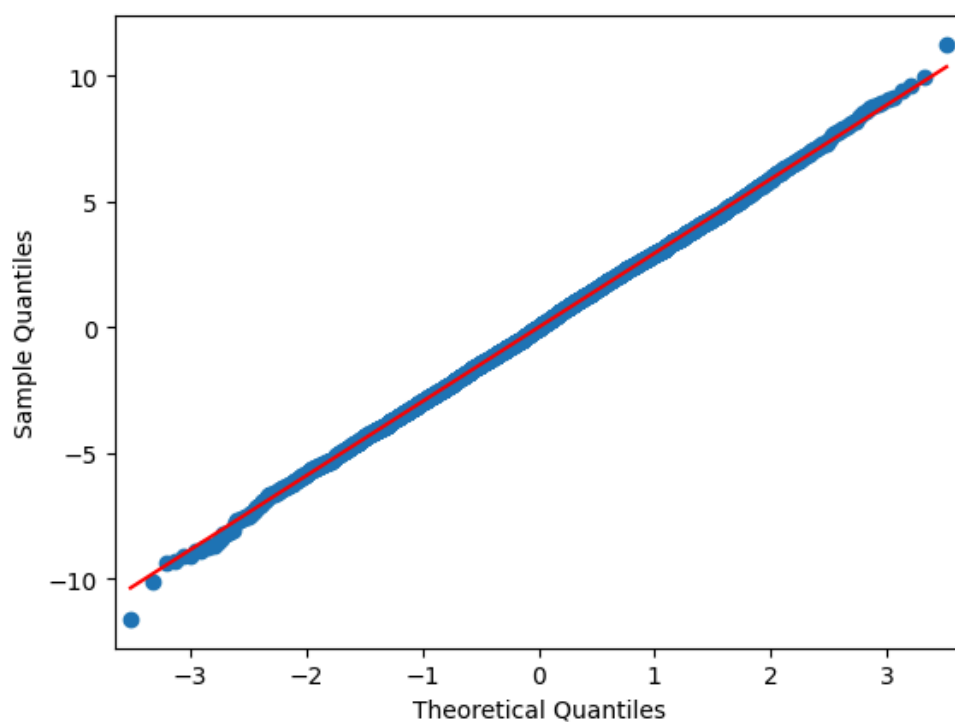




Another way to check the normality function is to create a quantile-quantile or Q-Q plot. If the residuals are normally distributed, you would expect a straight diagonal line going from the bottom left to the upper right of the Q-Q plot.

The Q-Q plot shows a similar pattern to the histogram, where the residuals are mostly normally distributed, except at the ends of the distribution.

```
import matplotlib.pyplot as plt
import statsmodels.api as sm
fig = sm.qqplot(model.resid, line = 's')
plt.show()
```



## ✓ Check the homoscedasticity assumption

Lastly, we have to check the homoscedasticity assumption. To check the homoscedasticity assumption, we can create a scatterplot of the fitted values and residuals. If the plot resembles a random cloud (i.e., the residuals are scattered randomly), then the assumption is likely met.

he first argument is the variable that goes on the x-axis. The second argument is the variable that goes on the y-axis.

```
import matplotlib.pyplot as plt
fig = sns.scatterplot(x=fitted_values, y=residuals)

# Add reference line at residuals = 0
fig.axhline(0)

# Set x-axis and y-axis labels
fig.set_xlabel("Fitted Values")
fig.set_ylabel("Residuals")

# Show the plot
```